

# NLP-Based Intelligent Web Application Development

Project Title: NLP Assistant - Intelligent Chatbot and Text Analyzer

## Overview:

This Flask web application processes human language through typed text or browser voice input. It returns chatbot responses, visible NLP analysis, speech output, and database-backed history.

## Technology Stack:

- Frontend: HTML, CSS, Bootstrap, JavaScript
- Backend: Python Flask
- NLP: NLTK with rule-based fallbacks
- Database: SQLite runtime database and SQL export file
- Speech: Browser Web Speech API and Speech Synthesis API

## Implemented NLP Concepts:

1. Tokenization - displays tokenized user input.
2. Sentiment Analysis - classifies Positive, Negative, or Neutral.
3. Voice-to-Text - microphone input converts speech to text.
4. Text-to-Speech - Speak button reads chatbot output aloud.
5. Chatbot System - user vs bot interface with stored history.
6. Keyword Extraction - highlights important words from input.
7. Text Classification - labels messages as Complaint, Inquiry, Feedback, Command, or General.

## System Workflow:

User input -> JavaScript fetch request -> Flask route -> NLP engine -> database save -> JSON response -> UI display

## Database Table: chats

Fields: id, user\_message, bot\_response, tokens, keywords, sentiment, sentiment\_score, classification, created\_at.

## How to Run:

```
python3 -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt
python3 app.py
Open http://127.0.0.1:5000
```

## Defense Demo Checklist:

Show chatbot, token list, sentiment result, keywords, classification, voice input, speech output, and history table.